St-CGN: Multiscale Consistency and Dual-View Entropy Fusion for Source-Free Domain Adaptation in Remote Sensing Semantic Segmentation

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***Abstract*—In unsupervised domain adaptation (UDA) for remote sensing (RS), source-free domain adaptation (SFDA) has emerged as a crucial paradigm to overcome strict data limitations. However, traditional self-training (ST) methods struggle with severe domain shifts in SFDA, which cause spatial and semantic inconsistencies that lead to early misclassifications into severe pseudolabel noise. To resolve these problems, we propose St-CGN, a novel two-stage framework designed to achieve robust class-level adaptation by establishing strict multiscale feature invariance and utilizing class-balanced recalibration to ensure pixel-level accuracy in the UDA mission of RS. First, to achieve better feature robustness and improve the model's ability to capture features with semantic correlations, a feature-consistency learning (CL) strategy is proposed, which sculpts a semantic-invariant feature space for the network, ensuring that varied visual representations of the same semantic content are mapped in a consistent manner. Second, a prototype feature guidance ST procedure based on a pseudolabel refining strategy is proposed to ease the uncertainty predictions, leveraging the semantic feature spaces to mitigate the class bias caused by domain shift. Extensive experiments conducted on widely used datasets verify the effectiveness of the proposed method for RS applications.**

***Index Terms*—Consistency learning (CL), pseudolabel recalibration, remote sensing images (RSIs), self-training (ST), source-free domain adaptation (SFDA).**

I. Introduction

OVER the past decades, significant advancements in remote sensing (RS) technology and interpretation methodologies have revolutionized the processing capabilities for high-resolution RS images (HRSIs), thereby substantially enhancing surface monitoring applications [1], [2]. In particular, deep learning-based semantic segmentation has demonstrated exceptional efficacy across a wide range of scenarios, such as land use and land-cover mapping. Consequently, these techniques play a crucial role in enabling intelligent analysis for informed decision-making [3], [4], [5].

While this progress has been largely driven by deep convolutional neural networks (DCNNs), which have achieved remarkable success when provided with sufficient annotations [6], [7], significant challenges persist. The performance of these models remains heavily reliant on extensive labeled datasets from the target domain, entailing prohibitive data curation costs and operational constraints. Furthermore, standard DCNN-based approaches exhibit high sensitivity to data distribution shifts and domain variance, posing a substantial barrier to meeting the demands of RS analysis [8], [9]. Consequently, it is essential to develop generalization of DCNNs to enable accurate cross-domain predictions and bridge distribution gaps without further annotations.

Domain adaptation methods have recently garnered significant attention for enhancing segmentation performance in cross-domain scenarios, primarily by mitigating the reliance on large-scale labeled data through unsupervised or weakly supervised techniques [10], [11], [12], [13], [14], [15]. Within this field, unsupervised domain adaptive semantic segmentation (UDASS) has emerged as a particularly promising paradigm, facilitating effective knowledge transfer across distinct domains. Current UDASS approaches generally follow two paradigms: adversarial learning (AL) and self-training (ST). Adversarial adaptive approaches aim to minimize adversarial losses at various training stages to reduce domain discrepancies. This is often accomplished by aligning features at different levels, such as modifying image styles, generating domain-invariant feature spaces, or aligning distributions in the prediction output space [14], [16]. However, when applied to complex scenes with substantial domain gaps, adversarial alignment frequently struggles to preserve fine-grained spatial structures. Consequently, ST has emerged as a superior paradigm for this task, iteratively refining the model using high-confidence target predictions to better preserve essential spatial fidelity [15], [17], [18].

Despite its structural advantages over AL, the ST paradigm remains fundamentally constrained by its strict dependency on pseudolabel quality, and since early spatial misclassifications can easily propagate and misguide the adaptation trajectory,

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several strategies have been proposed to mitigate the impact of label noise [19], [20], [21]. Prior works address bias through class-wise confidence [19] or category alignment [20]. A different approach by ProDA [22] utilized class prototypes to perform pseudolabel denoising based on feature distance. While these methods have significantly advanced the field, their underlying strategies suffer from two fundamental limitations when applied to the unique complexities of HRSIs: 1) unreliable pseudolabels severely degrade feature centroids in complex geographic imagery [23]; and 2) sampling strategies merely correct selection bias rather than dense label noise, failing to ensure pixel-level accuracy [15], [17]. High spectral similarity over land-cover categories causes confident misclassifications to amplify during training, accumulating massive semantic noise. Therefore, robust adaptation must bridge high-level semantic abstraction with the precise spatial fidelity required for dense targets.

To overcome these inherent limitations and successfully extract structured, robust features from two separate domains without solely relying on fragile pseudolabels or global alignments, consistency learning (CL) has emerged as a powerful paradigm [24], [25]. The objective of this strategy is to enforce representation invariance, forcing the model to maintain feature stability across augmented visual variations. By enforcing representation stability under perturbations, this approach mitigates overfitting to domain-specific cues in favor of learning intrinsic data structures. Existing methods generally target either high-level semantic consistency or low-level spatial consistency. For example, Silva and Rivera [25] propose learning representations by enforcing consistent cluster assignments that effectively align distinct image views toward the same set of prototypes. However, simply migrating these high-level semantic constraints to dense prediction tasks like semantic segmentation for HRSIs proves insufficient.

As we just mentioned with [26], despite the conceptual advancements of these unsupervised methods in natural image processing, simply adopting them for HRSIs is insufficient to resolve the cross-domain challenges inherent to RS [26], [27], [28], [29]. Compared to natural images, HRSIs exhibit substantial distributional deviations due to the unique imaging perspectives and the vast, heterogeneous geographical expanses captured by sensors [26]. Consequently, these images are characterized by heightened spatial complexity, immense resolution, and significant scale variance [27], [30], [31]. This frequently causes greater intraclass variance and interclass similarity. Furthermore, the extensive spatial dimensions of HRSIs require preprocessing steps such as cropping or downsampling to adapt the networks, which inevitably lead to information loss and boundary discontinuities, disrupting the spatial coherence of the features and degrading overall accuracy [30]. Consequently, building a robust adaptation framework to ease domain shift under RS scenario is essential, which can reconcile global scene-level context with fine-grained local details. Conventional global adversarial alignment and high-confidence pseudolabeling mechanisms frequently lack the granularity required to maintain consistency across these disparate spatial scales.

To resolve these inherent spatial and semantic limitations of UDASS for HRSIs, we propose St-CGN, a novel two-stage ST class-guided semantic segmentation network. As illustrated in Fig. 1, the framework initiates with a warm-up stage to construct a robust foundational feature space to provide reliable anchors for further recalibration. In the subsequent ST stage, to overcome the original vulnerabilities of HRSI UDASS missions, we mainly focused on pseudolabel corrections with a novel adaptive weighted class-balanced approach to ease the confusion of target predictions and suppress predictive variance by using entropy-based constraints, while using semantic as guidance for dense heterogeneous target predictions. Ultimately, this cohesive architecture guarantees stable predictions across diverse target domains. The main contributions of this work are summarized as follows.

1) We propose St-CGN, a novel two-stage source-free domain adaptation (SFDA) framework that fundamentally departs from previous methods. By establishing the semantic similarity within feature space and probability space, the framework prevents the accumulation of semantic noise across HRSIs domains. Further experiments on comprehensive datasets demonstrate the superior ability of St-CGN to overcome severe geographical and optical domain shifts in HRSIs.
2) We have introduced multiscale feature CL (MsFCL) and class-balance pseudolabel correction model (CbPCM) to address pseudolabel noise caused by domain shifts in HRSIs. During the warm-up feature learning stage, MsFCL constructs a robust feature space by establishing stable anchor prototypes through global instance-wise semantic representations and local region-wise spatial features. In the ST stage, CbPCM mitigates semantic noise by directly recalibrating target predictions against anchor prototypes to filter out low-confidence results for spectrally similar land-cover categories.
3) To resolve insufficient pixel-level accuracy and the loss of fine-grained spatial details, we introduce ASGCF and dual-view prediction entropy fusion (DvPEF) to maintain structural and spatial consistency. The ASGCF balances the discrepancy between high-level semantic context and low-level spatial details by employing semantic-based gating to effectively suppress background noise and preserve precise spatial boundaries, while DvPEF minimizes the entropy of multiview probability distributions, suppressing spatial instability to ensure consistent predictions across complex heterogeneous domains.

II. Related Work

A. UDA for RS Semantic Segmentation

Unsupervised domain adaptation (UDA) transfers knowledge from labeled source domains to unlabeled target domains. To achieve this, existing approaches can be categorized into AL and ST methods [10], [12], [32], [33]. In general semantic segmentation tasks, AL frameworks primarily focus on cross-domain distribution alignment [34]. For example,

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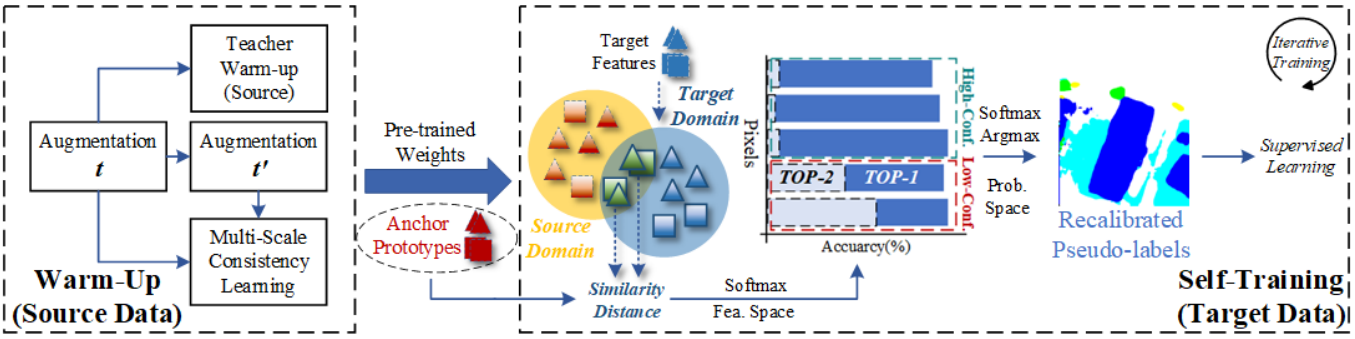


Fig. 1. Overall workflow of St-CGN. The framework is constructed into an ST stage with warm-up. The model establishes a semantic foundation and generates solid anchor prototypes over the source domain. The ST stage will leverage these anchors and source weights to gradually transfer knowledge of teacher network on source domain into the student network, ensuring that the framework progressively learns the target feature distribution with different characteristics on the probability space and feature space.

AdaptSeg [35] aligns the structured output space, while ADVENT [36] enforces domain invariance by minimizing prediction entropy. In comparison, ST approaches rely on target-generated pseudolabels for direct adaptation. Within this paradigm, DAFormer [21] established a robust foundation by integrating a Transformer architecture with rare class sampling to mitigate confirmation bias. Building upon this work, MIC [28] and HRDA [37] further enhanced adaptation through context-wise and mask consistency constraints. While successfully applied to natural images, standard UDA struggles with the complexity of HRSIs, where fluctuating sensor conditions and viewing angles create severe photometric and structural variations. In response, research has converged on AL and ST paradigms. For AL methods, such as BiFDANet [27], FSGAN [38], and SSC-CycleGAN [39], integrate CycleGAN [40] to align cross-domain representations. While FSGAN unifies global image styles to mitigate imaging variations, SSC-CycleGAN enforces dual-consistency constraints to address scale and style discrepancies. Furthermore, De-GLGAN [34] decouples feature maps into distinct frequency components while capturing cross-domain dependency relationships to achieve global–local alignments. Despite achieving domain invariance, adversarial methods often sacrifice semantic consistency, erode fine-grained spatial details, and introduce inherent optimization difficulties [41]. In contrast, ST offers an efficient way to exploit target-specific information by leveraging pseudolabel, which is generated directly from the target domain [31]. For RS, ST-DASegNet [42] applies ST to filter out domain-specific noise, while Li et al. [41] integrate mask-enhanced class mix (MECM) with scale-based rare class sampling (SRCS) to simultaneously alleviate class confusion and object scale variations in complex RS landscapes. However, ensuring pseudolabel accuracy and semantic integrity in HRSIs remains challenging.

B. CL in Domain Adaptation

In recent years, CL has emerged as a robust paradigm for learning representations from unlabeled data by minimizing feature divergence between differently augmented views. Foundational approaches like MoCo [12] and SimCLR [43] leverage instance-level pretext tasks to enforce consistency. By maximizing feature alignment between diverse augmented views, these approaches learn robust representations without semantic labels and are generally categorized by their level of consistency maintenance. At the input level, approaches like Lai et al. [44] and Olsson et al. [45] utilize advanced augmentation strategies such as context-aware perturbations and semantic mixing to promote prediction invariance. At the prediction level, methods aim to regularize the model by aligning outputs from different network branches. For example, Liu et al. [46] enforce consistency between student and teacher networks. To further stabilize the learning process, Xu et al. [47] minimize the divergence between parallel classifier heads, while C-SFDA [48] enforces consistent predictions across different training stages by employing a curriculum strategy to stabilize the optimization. Similarly, SJT-Net [30] utilizes spatial reconstruction and joint training within a transformer framework to ensure spatial and feature-level consistency across domains.

C. Noisy Label Correction

ST facilitates domain adaptation by iteratively generating pseudolabels for unlabeled target data to guide model refinement. However, as Zhang et al. [49] proposed, self-generated pseudolabels are inherently noisy, which leads to performance degradation by interfering with the learning process. This issue of noise has become a key challenge in ST methods, prompting the development of various strategies aimed at refining pseudolabels and minimizing the impact of this erroneous knowledge. For example, some methods, such as those proposed by Peng et al. [50], operate in the feature space by leveraging class prototypes to denoise pseudolabels. In contrast, other strategies introduced by CRST [51] and Pan et al. [52] function in the probability space aiming to directly correct label distributions through class-balance learning, while C-SFDA [48] utilizes selective pseudolabeling to progressively filter out noisy labels. Nevertheless, current approaches still leave high-confidence semantic noise and fundamental misalignments unaddressed.

III. Methodology

A. Preliminary

1) Problem Setups: The overall workflow of the proposed St-CGN is illustrated in Fig. 2. To effectively leverage

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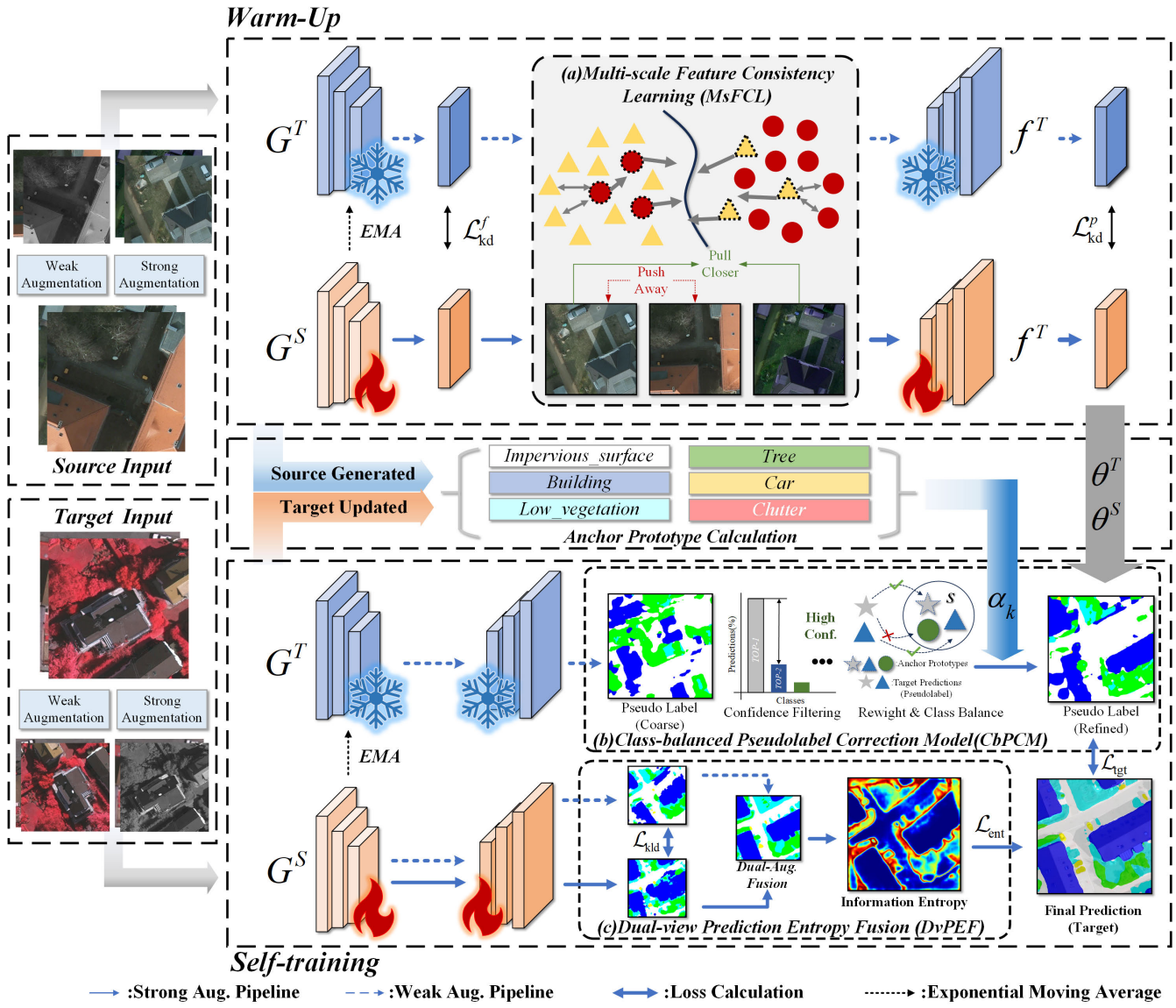


Fig. 2. Overall workflow of St-CGN. The warm-up stage focuses on establishing foundational models through MsFCL, creating a clean feature space for subsequent knowledge transfer, and utilizing knowledge distillation losses $\mathcal{L}_{\text{kd}}^{f}$ and $\mathcal{L}_{\text{kd}}^{p}$ over feature maps and predictions from the teacher and student backbones G^T and G^S, and decoder f^T and f^S. In the ST stage, pseudolabels are refined through CbPCM. Additionally, DvPEF processes two augmented views with KL-Divergence loss $\mathcal{L}_{\text{kld}}$, followed by information entropy regularization with $\mathcal{L}_{\text{ent}}$ for the final output.

knowledge from both the source and target domains, the framework operates across two distinct, weight-sharing stages. Formally, let $D_s = \{(x_s^i, y_s^i)\}_{i=1}^{N_s}$ denote the labeled source domain, where x_s represents the input and y_s represents the corresponding ground-truth label for C semantic categories. The target domain in the same way is defined as D_t, which shares the same label formats like the source domain. Under the constraints of UDA setting, the target labels remain inaccessible during the adaptation phase. As detailed in Section I, the discrepancy between these domains presents a significant barrier to direct model transfer.

2) Data Augmentation Strategy: For a given input x, we employ two independent data augmentation transformations, t and t', which are randomly sampled from an augmentation pool $\mathcal{T}$. They are applied to generate diversified views of the same instance. Specifically, the augmentation strategy follows the protocol established in [53], comprising random horizontal flipping, rotation, Gaussian blur, color jittering, and grayscale conversion.

3) Semantic Anchor Initialization: Finally, we define the set of anchor prototypes as $\mathcal{P}^{(0)} = \{p_k^{(0)}\}_{k=1}^{C}$ to bridge the discrepancy between the source and target domains. These prototypes capture the clean distribution statistics of the source domain and serve as the initialization for subsequent adaptation. Using the teacher network f_{θ_t} for its stability and knowledge from the source domain, we extract feature maps for all source images and apply masked global average pooling to compute the initial anchor $p_k^{(0)}$ for each class k

$$P_k^{(0)} = \frac{\sum_{x_s \in D_s} \sum_{i,j} \mathbb{1}\left[y_s^{(i,j)} = k\right] \cdot E_{\theta_t}(x_s)^{(i,j)}}{\sum_{x_s \in D_s} \sum_{i,j} \mathbb{1}\left[y_s^{(i,j)} = k\right]} \tag{1}$$

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where θ_t denotes the teacher network's weight. $\mathbb{1}[\cdot]$ denotes the indicator function, acting as a pixel-wise filter. It evaluates to 1 solely when the ground-truth label $y_s^{(i,j)}$ at spatial position (i, j) strictly belongs to category k and 0 otherwise. This masking mechanism ensures that the prototype calculation selectively aggregates only the relevant class-specific semantic features, effectively filtering out irrelevant background information.

B. Feature-Driven Semantic Foundation Learning

1) Instance-Wise Semantic CL: To establish a robust feature space preserving both semantic integrity and spatial sensitivity, we propose the MSFCL, as shown in Fig. 3. The framework achieves comprehensive representation learning by applying multiscale consistency constraints across both global instance-wise semantic representations and local region-wise spatial features.

A single HRSI typically exhibits significant semantic heterogeneity. Consequently, treating the entire image as a single instance can blur the distinction between spectral signatures and feature representations. To address this, we implement a class-aware masked pooling strategy designed to extract disentangled, class-specific prototypes for CL.

Formally, we define the semantic segmentation network as consisting of a feature encoder G and the decoder f. We adopt a teacher–student framework [54], where the teacher network's weights evolve as an exponential moving average (EMA) of the student network's weights and use α as the EMA coefficient

$$\theta_t = \alpha\theta_t + (1-\alpha)\,\theta_s \tag{2}$$

where θ_t and θ_s denote the weight of both teacher and student network. For a given input, let f_s and f_t represent the feature maps projected by the student and teacher encoders, respectively. Using the ground-truth mask Y, we identify the set of unique semantic categories, denoted as C_p, present in the current view. For each class $k \in C_p$, we generate a binary mask M_k, where $M_k^{(i,j)} = 1$ if the label at spatial position (i, j) corresponds to category k, and 0 otherwise. The class-specific prototype vector v_k is obtained by performing global average pooling over the features restricted to the masked region

$$v_k = \frac{\sum_{i=1}^{H}\sum_{j=1}^{W} M_k^{(i,j)} \cdot F^{(i,j)}}{\sum_{i=1}^{H}\sum_{j=1}^{W} M_k^{(i,j)}} \tag{3}$$

where $F^{(i,j)}$ denotes the feature vector at spatial position (i, j). The extracted prototypes are L2-normalized to yield the student and teacher class prototypes, z_k^s and z_k^t. To enforce semantic invariance, an InfoNCE-based contrastive loss maximizes the cosine similarity between these counterparts, where l iterates over all present categories in C_p to compute the normalization term. The consistency loss for a single input instance is averaged over all categories present in the view

$$\mathcal{L}_{\text{inst}} = -\frac{1}{|C_p|} \times \sum_{k\in C_p} \log \frac{\exp\left(sim\left(z_k^s, z_k^t\right)/\tau\right)}{\sum_{l\in C_P} \exp\left(sim\left(z_k^s, z_l^t\right)/\tau\right)} \tag{4}$$

where τ is the temperature hyperparameter. This mechanism ensures that semantic representations remain invariant to strong data augmentations, thereby enhancing intraclass consistency and encouraging the student network to map diverse augmented views of the same categories into compact, unified clusters within the feature space.

2) Region-Wise Spatial CL: While instance-wise consistency aligns representations at the global class level, it may overlook fine-grained spatial details essential for dense prediction tasks. To address this, we introduce a region-wise branch operating with spatial features that enforce feature invariance at the local level. Standard pixel-wise consistency methods typically assume a strict spatial correspondence between views [46], [47]. However, strong augmentations (e.g., random cropping and rotation) disrupt this spatial alignment, rendering direct coordinate-based matching ineffective. Although several strategies such as NCNet [55] and CoPDASeg [56] that ease coordinate dependencies have been proposed recently, they still face substantial computational burdens when applied to HRSIs and lack robust mechanisms to prevent erroneous matchings when facing severe augmentations.

To solve these, we propose a global matching and adaptive filtering strategy. Let F^t and $F^s \in \mathbb{R}^{C\times H\times W}$ denote the feature maps from the teacher and student projection heads, respectively. We flatten these maps into feature vectors $\mathcal{V}^t = \{\mathbf{v}_i^t\}_{i=1}^{HW}$ and $\mathcal{V}^s = \{\mathbf{v}_j^s\}_{j=1}^{HW}$. To maintain efficiency, we randomly sample a subset of anchors $\mathcal{A} \subset \mathcal{V}^t$ from the teacher's view, which operate as anchors for the following calculations. For each anchor, we identify its corresponding positive pair in the student view by searching for the pixel with the maximum cosine similarity across the entire student feature map. This effectively locates the same semantic region regardless of its spatial displacement. The matched positive sample v_{pos}^s is defined as

$$v_{pos}^s = \arg\max_{v_j^s\in\mathcal{V}^s} sim\left(v_i^t, v_j^s\right) \tag{5}$$

where $\text{sim}(\cdot,\cdot)$ denotes cosine similarity. To prevent the model from learning erroneous correlations from mismatched pairs (e.g., when the anchored region is entirely cropped out of the student view or objects being shifted from the view), we employ a dynamic increasing similarity threshold δ. Let δ^t denote the threshold at iteration t, initialized to an empirical baseline of $\delta^0 = 0.7$. An anchor is considered valid only if its maximum cosine similarity surpasses the current threshold $sim(v_i^t, v_{pos}^s) > \delta^0$.

To ensure that the threshold dynamically adapts to the evolving feature space, we introduce a curriculum-based strategy to progressively elevate the matching requirement as the student network's representation capability improves. We first identify the least confident similarity score among these valid matches, defined as s_{min}^t. At the end of the iteration, the updated threshold δ^t is determined by comparing this minimum valid similarity against the previous threshold δ^{t-1}, while bounded with an upper limit of $\delta_{max} = 0.96$

$$\delta^t = \min\left(\delta_{max}, \max\left(\delta^{t-1}, s_{min}^t\right)\right). \tag{6}$$

For each valid anchor $\mathbf{v}_i^t$, we compute the probability P_i of correctly identifying $\mathbf{v}_{pos}^s$ against a set of N_{neg} negative samples $\mathcal{N}_i$, which are randomly drawn from the remaining student

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features $\mathcal{V}^s \setminus \{\mathbf{v}^s_{pos}\}$. Specifically, we set $N_{neg} = 255$ to reduce memory use and accelerate the calculation

$$P_i = \frac{\exp\left(\text{sim}\left(\mathbf{v}^t_i, \mathbf{v}^s_{pos}\right)/\tau\right)}{\exp\left(\text{sim}\left(\mathbf{v}^t_i, \mathbf{v}^s_{pos}\right)/\tau\right) + \sum_{\mathbf{v}^s_k \in \mathcal{N}_i} \exp\left(\text{sim}\left(\mathbf{v}^t_i, \mathbf{v}^s_k\right)/\tau\right)}. \tag{7}$$

Consequently, the total region-wise consistency loss is computed by minimizing the contrastive objective averaged over all valid anchors

$$\mathcal{L}_{\text{reg}} = -\frac{1}{|\mathcal{A}'|} \sum_{\mathbf{v}^t_i \in \mathcal{A}'} \log\left(P_i\right). \tag{8}$$

This objective encourages the student network to maintain local feature consistency with the stable teacher network, ensuring that fine-grained texture and boundary information are preserved against domain-specific perturbations.

3) Overall Consistency Objective: The total objective of the MsFCL module integrates complementary instance- and region-level constraints, further augmented by knowledge distillation to ensure strict alignment between the student and teacher networks. The feature CL objective integrates two complementary constraints comprising the instance- and the region-level consistency. The combined objective is formulated as

$$\mathcal{L}_{\text{fcl}} = \lambda_{\text{inst}}\mathcal{L}_{\text{inst}} + \lambda_{\text{reg}}\mathcal{L}_{\text{reg}} \tag{9}$$

where $\mathcal{L}_{\text{inst}}$ and $\mathcal{L}_{\text{reg}}$ represent the instance-wise and region-wise consistency loss, respectively, and λ_{inst} and λ_{reg} represent the corresponding weights. Furthermore, to directly constrain the student network to align with the stable representations of the teacher, we introduce a distillation objective, which comprises two components consisting of the output-level distillation loss $\mathcal{L}^p_{\text{kd}}$ and the feature-level distillation loss $\mathcal{L}^f_{\text{kd}}$. Consequently, the final overall consistency objective combines these constraints to drive the student network toward learning representations that are categorically well-separated and aligned with the teacher's robust prototype space

$$\mathcal{L}_{\text{msfcl}} = \mathcal{L}_{\text{fcl}} + \mathcal{L}^p_{\text{kd}} + \mathcal{L}^f_{\text{kd}}. \tag{10}$$

C. Prototype-Guided Calibration With Dual-View Consistency

1) Class-Balanced Pseudolabel Correction Model: In SFDA, pseudolabels generated on the target domain are often corrupted by domain shift, with errors being particularly pronounced for visually confused or underrepresented classes. Conventional ST typically filters pseudolabels using a global softmax confidence threshold, which can exacerbate early misclassifications and induce confirmation bias.

In this framework, the source anchor prototypes serve as a stable semantic reference, representing the ideal feature distribution, while the student's prediction on the target data constitutes the observation evidence. Formally, let $\mathbf{z}_i$ denote the l_2-normalized feature vector of target pixel i, and let $\mathcal{P} = \{\mathbf{p}_k\}_{k=1}^C$ denote the set of source prototypes. We first compute the semantic evidence score $S_{i,k}$ by treating the cosine similarity between $\mathbf{z}_i$ and $\mathbf{p}_k$ as a proxy for class probability

$$S_{i,k} = \frac{\exp\left(\text{sim}\left(\mathbf{z}_i, \mathbf{p}_k\right)/\delta_{\text{cal}}\right)}{\sum_{j=1}^C \exp\left(\text{sim}\left(\mathbf{z}_i, \mathbf{p}_j\right)/\delta_{\text{cal}}\right)} \tag{11}$$

where δ_{cal} denotes the calibration temperature to better adjust the confidence scaling of the prototype similarity distribution. This similarity distribution functions as a semantic prior to rectify the model's predictive uncertainty. Specifically, the rectified probability distribution $\hat{\mathbf{P}}_i$ is obtained by performing a semantic-guided fusion, computed as the l_1-normalized element-wise product of the student's initial softmax probabilities $\mathbf{P}_i$ and the semantic weight $\mathbf{S}_i$. To ensure sufficient supervision for minority categories, we employ a dynamic class-balanced thresholding strategy. Specifically, we introduce a category-wise dynamic threshold β_k, defined as the probability score ranked at the top p_{ratio} proportion among the original student predictions $\mathbf{P}$ within each mini-batch. Using the original distribution for this threshold preserves statistical stability against domain shift. The p_{ratio} is gradually increased from the initial 0.4 to the final 0.9. This progressive scheduling is introduced to prevent the degradation of the model caused by noise during the beginning of domain adaptation. The model will learn strictly from high-confidence pseudolabels in the early stage. As optimization proceeds, the target feature space will gradually stabilize, and the threshold is further relaxed to capture more comprehensive target representations.

Finally, the pseudolabel $\hat{y}_i$ is selected from the rectified distribution $\hat{\mathbf{P}}_i$. To account for domain shift, this assignment is validated against the original prediction confidence. Supervision is applied only if the original probability meets the class-specific threshold

$$\hat{y}_i = \begin{cases} k^*, & \text{if } \mathbf{P}_{i,k^*} \geq \beta_{k^*} \\ \text{ignore}, & \text{otherwise} \end{cases} \tag{12}$$

where $k^* = \arg\max \hat{\mathbf{P}}_{i,k}$. This mechanism effectively filters out high-confidence yet semantically inconsistent predictions, preventing calibration distortions from overriding established manifold structures.

2) Dual-Viewed Prediction Entropy Fusion: To enhance the supervision from the CbPCM module, we introduce the DvPEF, an uncertainty regularizer that promotes predictive stability across domains. We propose a dual-stream unified distribution to address overfitting to noise in student predictions and the high-confidence bias of the teacher. This distribution reduces variance from random augmentations by averaging the softmax probabilities from the weakly augmented teacher view $P^t(y|x_w)$ and the strongly augmented student view $P^s(y|x_s)$

$$\overline{P}\left(y|x\right) = \frac{1}{2}\left(P^t\left(y|x_w\right) + P^s\left(y|x_s\right)\right). \tag{13}$$

To regularize the predictions from both the teacher and student models, we introduce the KL-Divergence loss, which aligns their feature distributions across views

$$\mathcal{L}_{\text{kld}} = \text{KL}\left(P^s\left(y|x_s\right), P^t\left(y|x_w\right)\right) \tag{14}$$

and this ensures that two networks produce consistent feature-level predictions, promoting stability during adaptation. To encourage decision boundaries to settle in low-density regions while preventing collapse, we regularize the global entropy of the unified distribution. In contrast to standard pixel-wise entropy minimization, which sharpens individual predictions, we maximize the entropy of the marginal class distribution

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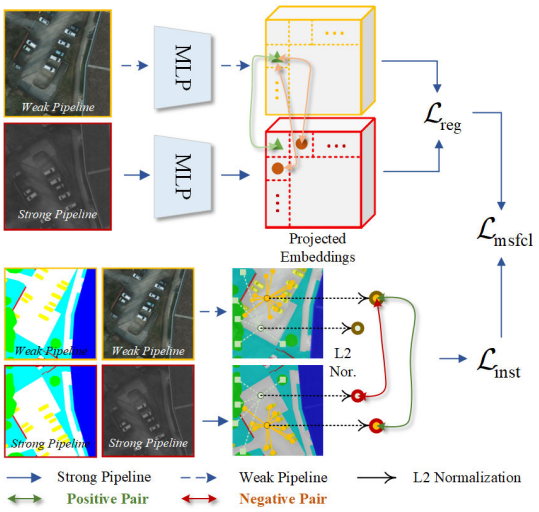


Fig. 3. Structure of the proposed MsFCL. MLPs act as projection heads to generate feature embeddings. These projected feature maps are subsequently used to enforce multiscale consistency across the network. The MLP is constructed using two 1×1 convolutional layers and a ReLU activation function.

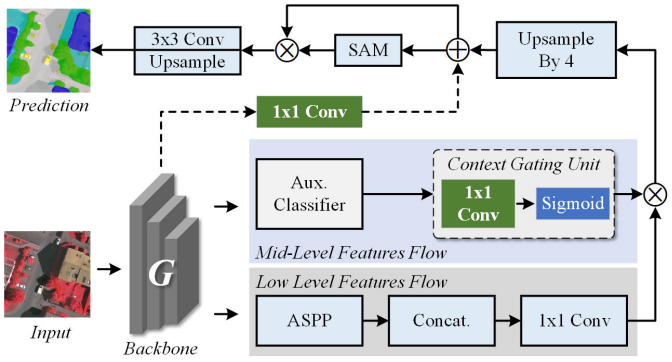


Fig. 4. Structure of the proposed ASGCF. The module leverages auxiliary logits to gate high-level features, fuses them with low-level spatial details, and refines the representation via the SAM to generate the final prediction. The low-level feature branch follows the standard DeepLabV3 + architecture.

over the mini-batch $\mathcal{B}$. This formulation implicitly promotes class diversity across samples. The marginal probability for category k, denoted as $\hat{p}_k$, is computed by averaging the unified probabilities across all spatial locations (h, w) and batch instances

$$\hat{p}_k = \frac{1}{|\mathcal{B}| \cdot H \cdot W} \sum_{x \in \mathcal{B}} \sum_{h,w} \overline{P}_k (y|x)^{(h,w)} \tag{15}$$

and the entropy regularization loss is defined to maximize the information content of this marginal distribution

$$\mathcal{L}_{\text{ent}} = \sum_{k=1}^{C} \hat{p}_c \log (\hat{p}_c) . \tag{16}$$

Consequently, the ST stage integrates the recalibrated pseudolabel supervision with this entropy regularization

$$\mathcal{L}_{\text{tgt}} = -\frac{1}{|\mathcal{B}| \cdot H \cdot W} \sum_{x \in \mathcal{B}} \sum_{i,j} M_{i,j} \log \left(P^{s,(i,j)}_{\hat{y}_{i,j}} \right) \tag{17}$$

$$\mathcal{L}_{\text{st}} = \mathcal{L}_{\text{tgt}} (\hat{y}, P^s) + \lambda_{\text{ent}} \mathcal{L}_{\text{ent}} + \lambda_{\text{kld}} \mathcal{L}_{\text{kld}} \tag{18}$$

where λ_{ent} and λ_{kld} denote the weighting coefficients for entropy regularization and consistent feature-level predictions, respectively, with $\lambda_{\text{kld}} = 1 - \lambda_{\text{ent}}$.

D. Auxiliary Semantic-Guided Context Fusion

As illustrated in Fig. 4, ASGCF mitigates the inherent discrepancy between high-level semantic abstraction and low-level spatial precision. Structures like DANet [57] and OCRNet [58] enhance representations by aggregating global or multiscale contexts within a single semantic representation space. However, directly fusing these abstract features with low-level spatial maps often blurs fine boundaries in HRSIs. To resolve this discrepancy, our proposed ASGCF utilizes auxiliary predictions to generate a spatial gating signal that actively filters out uncertain predictions and suppresses background noise to preserve sharp object boundaries and structural integrity.

Specifically, let $F_{\text{high}} \in \mathbb{R}^{C \times H \times W}$ denote the high-level semantic features and $\mathbf{Z}_{aux} \in \mathbb{R}^{K \times H \times W}$ denote the auxiliary logits. The context gating unit extracts a semantic gating signal from the auxiliary output by applying a 1×1 convolution followed by a Sigmoid activation

$$\mathcal{G} = \sigma \left(\text{Conv}_{1 \times 1} (\mathbf{Z}_{aux}) \right) . \tag{19}$$

To prevent signal degradation when gating weights are near zero, we employ a residual gating strategy where the high-level features are enhanced by adding the gated signal to the identity mapping. This can be defined as

$$F_{\text{gate}} = F_{\text{high}} + \left(F_{\text{high}} \odot \mathcal{G} \right) \tag{20}$$

where $\odot$ denotes element-wise multiplication. This ensures that the model focuses on regions with high semantic certainty while preserving foundational feature information. Following this semantic filtering, we fuse the gated features with mid-level backbone features F_{mid} to recover spatial resolution lost during downsampling. Both streams are projected to a unified channel dimension via independent 1×1 convolutions and aligned via bilinear interpolation. This results in a fused map F_{sum} that integrates the spatial texture of the backbone with the semantic context of the decoder

$$F_{\text{sum}} = \phi_{\text{mid}} (F_{\text{mid}}) + \phi_{high} \left(F_{\text{gate}} \right) . \tag{21}$$

To sharpen object boundaries, we employ a spatial attention module (SAM). This module first aggregates spatial information by concatenating the outputs of parallel channel-wise average and max pooling operations, yielding the spatial descriptor F_{pool}. Subsequently, this descriptor is processed by a large-kernel 7×7 convolution followed by a Sigmoid activation to infer the spatial attention map M_s

$$M_s = \sigma \left(\text{Conv}_{7 \times 7} \left(F_{\text{pool}} \right) \right) . \tag{22}$$

The learned attention map is then applied to the fused features through element-wise multiplication to suppress task-irrelevant regions, generating F_{ref}. Finally, these refined

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features are sent into the prediction head, consisting of a 3×3 refinement convolution followed by a 1×1 convolution to map the refined features to the C class logits. These low-resolution logits are then bilinearly upsampled to the original image resolution to yield the final prediction Y_{pred}

$$Y_{\text{pred}} = \mathcal{U}\left(\text{Conv}_{1\times1}\left(\text{ReLU}\left(\text{Conv}_{3\times3}\left(F_{\text{ref}}\right)\right)\right)\right) \tag{23}$$

where $\mathcal{U}$ denotes the bilinear interpolation operation that aligns the output with the input resolution.

E. Overall Optimization

The optimization of St-CGN proceeds in two distinct stages comprising a source-supervised warm-up and a source-free ST adaptation. In the preliminary warm-up phase, the model is trained on the source domain to establish a robust semantic foundation. We employ the standard pixel-wise cross-entropy loss to optimize the network using source ground-truth annotations

$$\mathcal{L}_{\text{src}} = -\frac{1}{H \times W} \sum_{n=1}^{H\times W} \sum_{c=1}^{C} y_{n,c}^{s} \log\left(p_{n,c}^{s}\right) \tag{24}$$

where $H \times W$ denotes the spatial dimensions of the input, C represents the number of classes, $p_{i,c}^{s} \in [0,1]$ is the predicted probability of pixel i belonging to class k, and $y_{i,c}^{s} \in \{0,1\}$ is the one-hot encoded source ground truth. Consequently, the total objective for the warm-up stage augments this supervised loss with the MsFCL regularization

$$\mathcal{L}_{\text{wp}} = \mathcal{L}_{\text{src}} + \mathcal{L}_{\text{msfcl}}. \tag{25}$$

In the subsequent domain adaptation stage, supervision is derived exclusively from the refined pseudolabels, which is explained in Section III-C. Finally, the overall optimization objective for the proposed St-CGN framework can be defined as

$$\mathcal{L}_{\text{st-cgn}} = \mathcal{L}_{\text{wp}} + \mathcal{L}_{\text{st}}. \tag{26}$$

IV. Experiments

A. Datasets

1) ISPRS Potsdam and Vaihingen Dataset: Experiments were conducted on the ISPRS Potsdam and Vaihingen datasets, which share six semantic categories, as shown in Table II. Potsdam provides 38 tiles in RGB and IRRG, whereas Vaihingen provides 33 tiles in IRRG. We processed the datasets by cropping them into 512×512 patches, generating 5472 images for Potsdam and 742 for Vaihingen.

2) LoveDA Urban and Rural Dataset: Experiments were conducted on the LoveDA dataset [59], which includes seven semantic categories, as shown in Table III. The urban and rural domains provide 1833 and 2358 tiles, respectively. We processed the dataset by cropping images into 512×512 patches, prioritizing the cross-domain scenarios of urban to rural and rural to urban.

B. Evaluation Metrics

To measure the performance of our model, we introduced the mean intersection over union (mIoU) as a generalized standard, while with the $F1$-score, precision, and recall as ablation study indicators to better visualize the model's ability. The calculation formulations are described as follows:

$$\text{mIoU} = \frac{1}{C} \sum_{k=1}^{C} \frac{\text{TP}}{\text{TP} + \text{FP} + \text{FN}} \tag{27}$$

$$\text{Recall} = \frac{\text{TP}}{\text{FN} + \text{TP}} \tag{28}$$

$$\text{Precision} = \frac{\text{TP}}{\text{FP} + \text{TP}} \tag{29}$$

$$\text{F1} = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \tag{30}$$

where TP, FP, and FN are the numbers of true positives, false positives, and false negatives, and C is the number of feature classes.

C. Experiment Setups

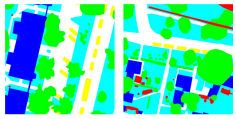
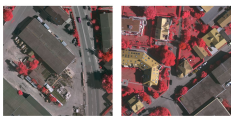
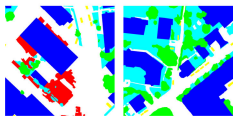
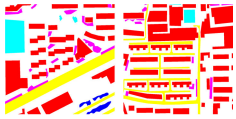
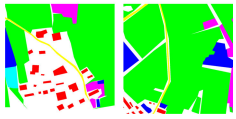
For a fair comparison, we evaluated several competitive methods under the same experimental configurations. These include AL methods such as Advent [36], AdaptSeg [35], and De-GLGAN [34], the ST methods, including C-SFDA [48], CRST [51], ST-DASegNet [42], SJT-Net [30], DAFormer [21], and MIC [28]. All experiments were conducted on a single NVIDIA GeForce RTX 4090. We utilize DeepLabV3+ [60] with a ResNet [61] backbone. The batch size is 4 and we use stochastic gradient descent for optimization with a learning rate of 0.01 and a weight decay of 0.0001. The total training consisted of 10 000 iterations, which were divided into three phases to ensure stable adaptation. The first 3000 iterations are dedicated to pretraining the teacher network on the source domain. The subsequent 2000 iterations focus on warming up the student network. The final 5000 iterations constitute the ST stage. Following the teacher's warm-up, the initial anchor prototypes are generated. During the ST phase, both prototypes and pseudolabels are updated every 1000 iterations to refine the model's adaptation from a source toward the target distribution. For data augmentation, the images from the LoveDA and ISPRS datasets were cropped into a size of 512×512 with augmentations.

D. Comparisons With Other Methods

1) Comparison Over ISPRS: The evaluation is conducted on two cross-domain scenarios within the ISPRS. The first task performs adaptation from the Potsdam dataset to the Vaihingen IRRG dataset, utilizing 4598 labeled source images and 543 unlabeled target images. Quantitative performance is assessed on the standard Vaihingen validation set comprising 149 images. Conversely, the second task utilizes the Vaihingen IRRG dataset as the source domain and the Potsdam RGB dataset as the target. This configuration leverages 543 labeled source images and 4598 unlabeled target images, with evaluation conducted on the Potsdam RGB validation set comprising 874 images.

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TABLE I
SPECIFICATIONS OF THE BENCHMARK DATASETS UTILIZED IN THE EXPERIMENTS

Datasets	Image	Label	Spatial Resolution	Spectral Bands	Category
ISPRS Potsdam			5 cm	RGB IRRG	Imp. surface Building Low vegetation Tree Car Clutter
ISPRS Vaihingen			9 cm	IRRG	
LoveDA Urban			30 cm	RGB	Background Building Road Water Barren Forest Agricultural
LoveDA Rural					

Quantitative Comparison: As presented in Table II, the framework was evaluated under cross-domain restrictions. The source-only baselines, utilizing ResNet-50 without ST, yielded reference mean IoU scores of 29.38% and 16.56% across two specific task settings. Our proposed method achieved substantial improvements, reaching 55.05% and 49.64% with ResNet-50, and further elevating to 55.34% and 49.83% with ResNet-101. These results significantly outperform the baseline and surpass the majority of prior state-of-the-art methods. Notably, the ASGCF module demonstrates distinct advantages for objects with complex boundaries or limited spatial extent. For instance, in the ResNet-50 Potsdam RGB to Vaihingen IRRG scenario, the car and tree categories exhibited gains of 17.65% and 30.01%, respectively. These enhancements are attributable to the proposed gating strategies, which effectively suppress background noise while mitigating the discrepancy between high-level semantic context and low-level spatial precision. Small objects like cars and irregular entities like trees are often overwhelmed by background clutter in high-resolution imagery. The gating mechanism addresses this by filtering irrelevant features, effectively preventing categorical confusion and ensuring robust segmentation of these target instances.

Furthermore, we introduce Ours(-) as an additional configuration that excludes the DvPEF and ASGCF modules to isolate and evaluate the core ST adaptation pipeline. After removing these two modules, we can still observe the performance improvement on the majority of semantic classes compared to the source-only. However, an invalid or ineffective transfer remains evident in more challenging categories, specifically car and clutter. We believe that this limitation is primarily driven by the class imbalance and long-tailed distribution of the HRSIs; the extreme scarcity of minority samples leads to insufficient representation learning in the framework. Consequently, Ours(-) on difficult classes also supports the contributions of both ASGCF and DvPEF, confirming their necessity in stabilizing prediction uncertainty and refining structural boundaries when facing limited supervision.

Qualitative Visual Results: As shown in Fig. 5(c), the source-only baseline exhibits significant categorical confusion and noisy predictions, reflecting a common failure mode in existing state-of-the-art methods. In contrast, our approach mitigates these distortions by establishing representation invariance across varying spatial resolutions, constructing a feature space impervious to geometric perturbations and scale shifts. Specifically, the model utilizes adaptive gating mechanisms to distinguish target features from irrelevant background clutter, thereby preserving spatial precision. As evidenced by Rows 4 and 10 in Fig. 5, this approach significantly suppresses noise and sharpens object boundaries, resulting in segmentation maps of superior structural integrity.

2) Comparsions Over LoveDA: We further evaluate the model on the LoveDA dataset to test robustness against geographical domain shifts. The urban-to-rural task employs 1156 labeled source and 1366 unlabeled target images, with validation performed on 992 rural images. The reverse task utilizes 1366 labeled rural source images and 1156 unlabeled urban target images, with evaluation conducted on the 677-image urban validation set.

Quantitative Comparison: A challenge of LoveDA is high spectral similarity between agriculture and forest categories, which often leads to confusion in standard adaptation methods. For instance, in the urban-to-rural task, the state-of-the-art MIC method struggles significantly, achieving only 21.81% IoU for agriculture. In contrast, St-CGN demonstrates exceptional discriminative capability in these spectrally complex scenarios, attaining 46.85% IoU for agriculture and 31.57%

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TABLE II

COMPARATIVE RESULTS OF CROSS-DOMAIN EXPERIMENTS ON ISPRS POTSDAM AND VAIHINGEN. THE HIGHEST RESULTS FOR EACH CATEGORY ARE HIGHLIGHTED IN BOLD. ALL EXPERIMENTS WERE CONDUCTED WITH RESNET50/101 BACKBONE FOR FAIR COMPARISON

Task	Methods	Type	Backbone	Per-class IoU (%)						mIoU (%)
				Imp.	Building	Low Veg.	Tree	Car	Clutter	
P_RGB - V_IRRG	Source only	-	ResNet50	56.12	23.67	12.16	52.61	25.32	6.70	29.38
	Advent [36]	AL		48.40	52.88	43.56	27.26	39.72	12.10	37.32
	SJT-Net [30]	ST		61.20	64.83	32.84	62.98	44.68	**50.42**	52.83
	AdaptSeg [35]	AL		59.48	72.11	33.39	57.74	17.37	8.02	41.35
	C-SFDA [48]	ST		62.34	74.52	45.38	70.23	13.43	45.71	51.94
	ST-DASegNet [42]	ST		67.12	75.32	47.43	55.13	43.90	20.11	51.50
	DAFormer [21]	ST		67.32	71.96	58.57	38.79	24.04	46.34	51.17
	MIC [28]	ST		70.13	**82.24**	42.98	58.55	**49.81**	17.55	53.54
	Ours(-)	ST		69.23	73.51	60.23	63.07	12.83	3.96	47.14
	Ours	ST		**71.16**	74.53	**66.43**	**70.72**	30.48	16.97	**55.05**
	ST-DASegNet [42]	ST	ResNet101	64.30	**75.91**	46.24	54.90	43.11	21.63	51.02
	DE-GLGAN [34]	AL		67.48	69.45	34.71	54.04	**49.83**	18.97	49.08
	C-SFDA [48]	ST		63.47	72.09	48.45	71.06	12.18	46.22	52.25
	DAFormer [21]	ST		67.39	71.87	59.71	37.28	28.81	**48.83**	52.32
	Ours	ST		**70.88**	75.50	**67.06**	**71.07**	30.07	17.43	**55.34**
V_IRRG - P_RGB	Source only	-	ResNet50	31.78	29.03	10.31	5.64	21.07	1.54	16.56
	Advent [36]	AL		37.47	54.33	15.46	30.67	42.29	6.14	31.06
	SJT-Net [30]	ST		58.67	65.17	51.28	**50.43**	43.15	28.34	49.51
	AdaptSeg [35]	AL		53.41	46.54	28.12	9.94	64.54	1.12	33.95
	C-SFDA [48]	ST		40.90	33.02	21.37	25.03	49.06	2.20	28.60
	ST-DASegNet [42]	ST		53.41	46.54	28.12	23.61	44.30	2.12	33.02
	DAFormer [21]	ST		19.79	50.05	52.30	5.92	25.36	31.10	30.75
	MIC [28]	ST		66.73	**76.15**	7.03	31.12	**72.65**	13.49	44.53
	Ours(-)	ST		43.18	55.67	42.71	24.10	23.39	15.84	34.15
	Ours	ST		**67.09**	62.17	**53.49**	31.82	52.18	**31.07**	**49.64**
	ST-DASegNet [42]	ST	ResNet101	56.42	46.54	28.12	23.61	44.30	2.12	33.02
	DE-GLGAN [34]	AL		61.51	**69.47**	22.93	**61.92**	25.89	17.21	43.15
	SJT-Net [30]	ST		60.44	65.49	50.98	51.00	41.86	29.40	**49.87**
	C-SFDA [48]	ST		41.95	34.17	21.37	26.11	49.89	3.16	29.44
	Ours	ST		**66.92**	61.80	**53.97**	32.05	**53.68**	**30.61**	49.83

for forest. These results surpass the previous SOTA MIC by margins of 25.04% and 4.74%. We attribute these substantial gains to the MsFCL mechanism. By establishing strict semantic invariance across strongly augmented views during the warm-up stage, the model learns to identify fundamental textural discrepancies between the structured patterns of agricultural landscapes and the irregular density of forests, rather than relying solely on color information. Furthermore, the ASGCF module maintains the spatial continuity of these large-scale agricultural regions, effectively preventing the fragmentation often observed in the prediction maps of competing methods.

Qualitative Visual Results: Fig. 6 illustrates the qualitative segmentation results for the LoveDA adaptation tasks. A characteristic of this dataset is the prevalence of complex, undefined semantic regions often categorized as background, which possess high semantic heterogeneity and occupy substantial areas. This complexity poses a substantial obstacle for conventional methods, as the feature overlap between these undefined regions and specific target classes frequently induces semantic confusion.

Consequently, the alternative approaches illustrated in Fig. 6(c)–(d) exhibit significant spatial incoherence, with fragmented predictions disrupting the semantic regions. However,

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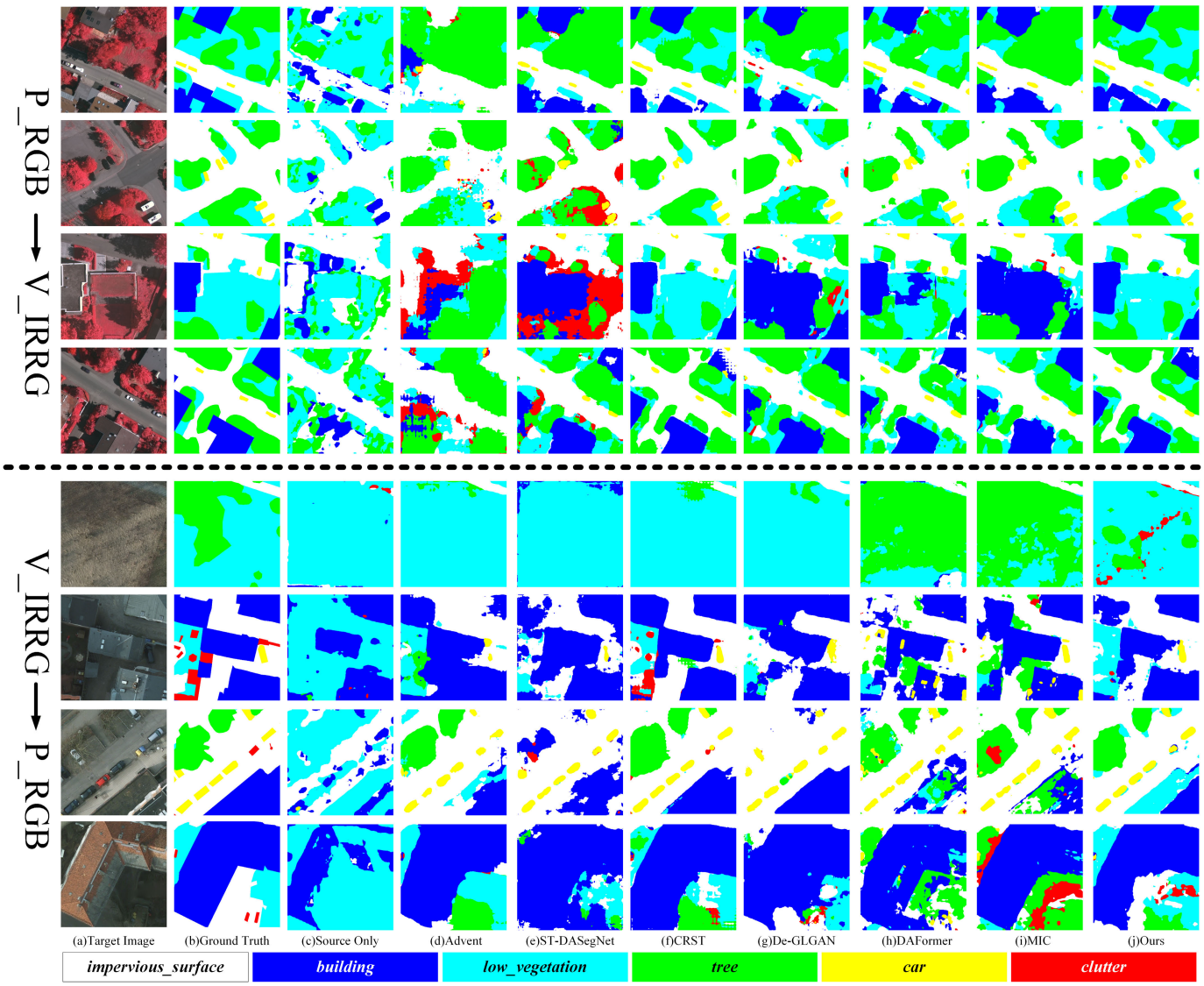


Fig. 5. Visualized prediction results over the ISPRS dataset. The original image, ground-truth labels, source-only, Advent [36], ST-DASegNet [42], CRST [51], De-GLGAN [34], DAFormer [21], MIC [28], and ours are from left to right.

St-CGN demonstrates robustness against these challenges. By utilizing the MsFCL mechanism, the framework encourages the learning of invariant features, mitigating noise induced by interclass resemblance. As a result, St-CGN generates segmentation masks with high intraclass consistency and reduced spatial fragmentation.

E. Visual Analysis of Semantic Consistency

As we mentioned in Section I, HRSI is inherently characterized by semantic complexity. In urban environments, spatially adjacent objects often exhibit high spectral and textural similarity, leading to significant feature confusion. As illustrated in Fig. 7(b), the model relying solely on pseudolabel correction suffers from severe spatial discontinuity. Corresponding high-entropy areas in the heatmaps expose uncertainty caused by visual similarity, specifically where the clutter category is wrongly predicted on building surfaces, producing fragmented prediction maps. Incorporating the ASGCF module in Fig. 7(c) partially mitigates fine-scale inconsistencies and sharpens object boundaries using semantic gating. In the absence of the semantic constraints imposed by MsFCL, the underlying feature representation continues to be influenced by domain-specific textures, causing misclassifications despite enhanced edge precision. Conversely, as shown in Fig. 7(d), the integration of entropy regularization significantly stabilizes predictions in high-uncertainty regions.

F. Ablation Experiment and Analysis

As detailed in Table IV, we conducted a stage-wise ablation study by decomposing St-CGN into three distinct configurations under the Potsdam RGB to Vaihingen IRRG task. Using warmup-only as a baseline for all groups, we assessed the isolated impact of our ASGCF module. The results demonstrate its strong generalization capability, achieving a 3.42% gain in mIoU. Moreover, combining ASGCF with the full framework boosts the mIoU from 51.72% to 55.05%. This

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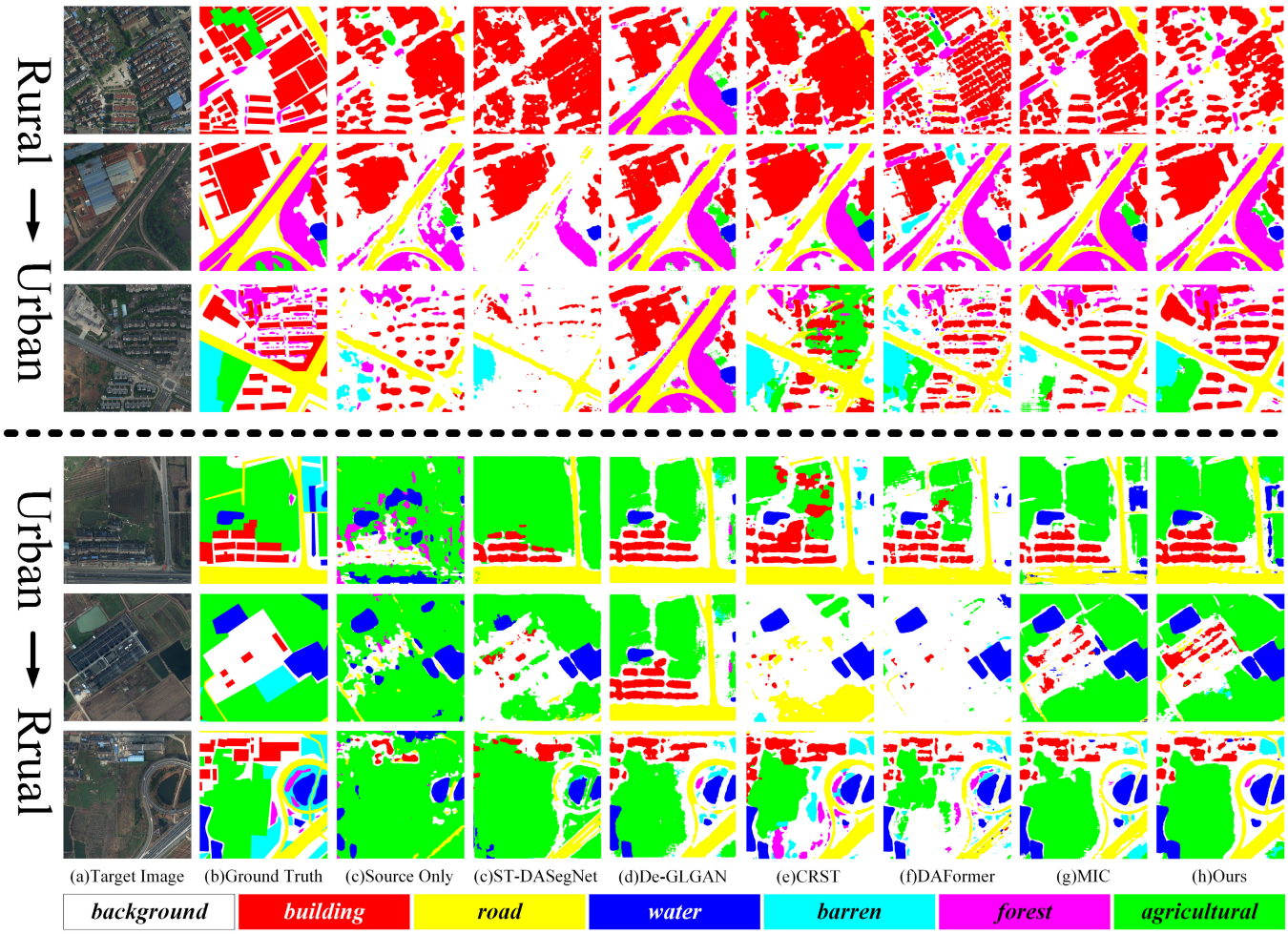


Fig. 6. Visualized prediction results over the LoveDA Dataset. The original image, ground-truth labels, source-only, ST-DASegNet [42], De-GLGAN [34], CRST [51], DAFormer [21], MIC [28], and ours are from left to right.

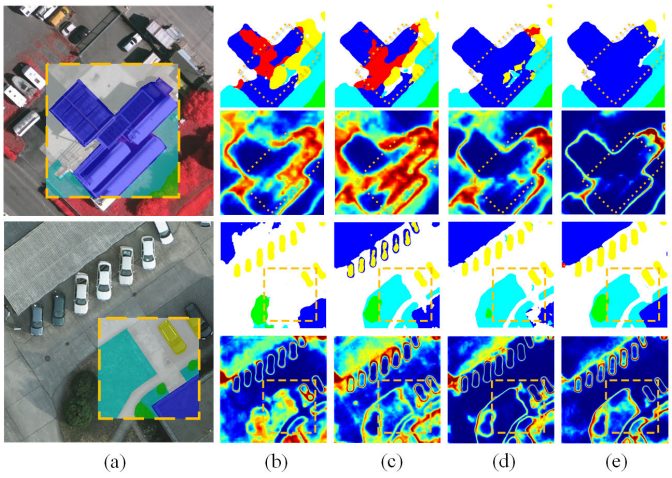


Fig. 7. Semantic segmentation results over different methods. (a) Original image with ground-truth label. (b) Warmup+CbPCM. (c) +ASGCF. (d) +DvPEF+MsFCL. (e) +DvPEF+MsFCL+ASGCF.

demonstrates that explicit guidance is essential for translating the feature foundation with proper guidance in deeper semantic space. Additionally, MsFCL enhances semantic understanding, enforcing a comprehensive and accurate representation across classes with diverse characteristics. By promoting cross-view feature invariance, this module strengthens the model's robustness to domain shift. In the initial warm-up stage, this approach achieves a 7.14% improvement in mIoU and a 5.24% boost in $F1$-score. This confirms that a stable feature space is crucial for effective knowledge transfer, as demonstrated by the significant disparity in results between the initialized and uninitialized models.

Moreover, the ST stage further enhances the prediction over the target domain. We performed ablation studies on two component configurations. When using only CbPCM, the results show limited gains compared to the combined CbPCM and DvPEF. This highlights the uncertainty of the proposed methods in cross-domain predictions based on anchor-driven refinement. While CbPCM generates more reliable pseudolabels, the network still suffers from noisy inputs and an unstable feature space. The DvPEF mechanism mitigates this, ensuring stability for both the teacher and student networks, thus improving the model's performance with CbPCM alone from 44.31% to 49.22%.

G. Comparison Over Different Imaging Condition

To evaluate the model's robustness against variations in imaging modalities, we conducted a comparative analysis

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TABLE III

COMPARATIVE RESULTS OF CROSS-DOMAIN EXPERIMENTS ON LOVEDA URBAN AND RURAL. THE HIGHEST RESULTS FOR EACH CATEGORY ARE HIGHLIGHTED IN BOLD. ALL EXPERIMENTS WERE CONDUCTED WITH RESNET-101 BACKBONE FOR FAIR COMPARISON

Task	Methods	Type	Per-class IoU (%)							mIoU (%)
			Background	Building	Road	Water	Barren	Forest	Agriculture	
Rural - Urban	Source only	-	45.60	29.48	28.90	31.05	16.88	13.11	40.82	25.69
	SJT-Net [30]	ST	53.15	**63.47**	44.31	48.29	31.22	41.26	25.46	43.97
	AdaptSeg [35]	AL	48.62	30.41	41.22	40.01	29.18	38.65	28.12	36.60
	DE-GLGAN [34]	AL	50.03	41.65	42.99	68.02	36.61	35.90	**43.17**	45.48
	C-SFDA [48]	ST	41.65	56.01	50.22	45.48	44.82	50.66	31.81	45.81
	CRST [51]	ST	49.57	57.37	**58.82**	68.31	24.65	37.42	33.43	47.91
	ST-DASegNet [42]	ST	50.28	43.21	38.11	63.18	42.25	50.21	40.27	46.79
	DAFormer [21]	ST	**59.57**	49.64	51.97	55.70	54.13	35.03	34.68	48.67
	MIC [28]	ST	56.74	39.61	52.14	43.44	**64.47**	19.15	42.94	45.50
	Ours	ST	59.12	46.02	37.02	**69.18**	39.25	**50.87**	41.75	**49.03**
Urban - Rural	Source only	-	20.97	17.73	26.43	38.01	14.25	16.02	28.36	23.11
	SJT-Net [30]	ST	45.28	34.12	36.45	40.30	20.51	11.89	29.06	31.09
	AdaptSeg [35]	AL	41.35	29.89	40.61	40.92	46.62	19.70	31.05	35.73
	DE-GLGAN [34]	AL	45.31	49.57	38.73	48.63	25.49	29.31	46.23	40.46
	C-SFDA [48]	ST	35.48	47.89	36.26	**66.92**	28.14	9.43	41.98	38.01
	CRST [51]	ST	39.16	50.57	34.40	63.24	14.83	18.52	45.78	38.07
	ST-DASegNet [42]	ST	46.78	36.04	36.11	43.83	22.90	23.52	44.28	36.21
	DAFormer [21]	ST	40.68	38.64	**50.13**	40.93	**51.71**	18.27	40.34	40.10
	MIC [28]	ST	**53.26**	**51.45**	36.32	43.27	50.10	26.83	21.81	40.43
	Ours	ST	46.52	50.69	30.17	53.28	32.95	**31.57**	**46.85**	**41.72**

TABLE IV

ABLATION STUDY OVER DIFFERENT STAGES UNDER POTSDAM RGB TO VAIHINGEN IRRG TASK

Stage	MsFCL	ASGCF	CbPCM	DvPEF	mIoU(%)	F1(%)	Prec.(%)	Recall(%)
Warmup	×	×	×	×	29.38	45.12	65.01	44.35
	×	✓	×	×	32.69	52.34	67.49	50.66
	✓	✓	×	×	39.83	57.58	69.17	52.90
Self-Train (w/o Warmup)	×	×	✓	×	44.31	54.65	69.34	55.87
	×	×	×	✓	46.50	56.23	70.95	58.39
	×	×	✓	✓	49.22	61.63	69.81	59.46
Self-Train (w/ Warmup)	✓	×	✓	✓	51.72	65.30	69.68	67.03
	✓	✓	✓	✓	**55.05**	**71.39**	**68.47**	**70.88**

on the ISPRS dataset under different spectral configurations. As shown in Table V, the domain gap differs across source modalities. Competing methods, such as SJT-Net and DAFormer, suffer performance declines of 2.06% and 1.89%, respectively, when transitioning from Potsdam RGB to IRRG. Conversely, St-CGN maintains consistent performance with a marginal decline of just 0.12%, validating our design's effectiveness. Specifically, MsFCL facilitates this process by applying photometric perturbations to the inputs that simulate spectral shifts. This compels the model to develop robust structural representations, ensuring that feature extraction is based on object geometry and semantic identity. This robust foundation is further reinforced during the adaptation stage by CbPCM and DvPEF. Finally, ASGCF enhances the prediction by utilizing auxiliary semantic guidance to regulate the fusion of contextual information. This approach allows the model to effectively separate invariant semantic content from the varying spectral characteristics of the source data.

H. Hyperparameter Sensitivity Analysis

1) Instance-Wise and Region-Wise Consistency Weight: During the warm-up stage, the MsFCL establishes a robust feature space for domain adaptation. To balance global and local constraints, λ_{inst} controls the weight of instance-level consistency, while the region-level weight is defined as $\lambda_{\text{reg}} = 1 - \lambda_{\text{inst}}$. As shown in Fig. 8(a), by giving a lower weight for instance level, we achieve more focus and better results in overall prediction.

2) Entropy Regularization: The hyperparameter λ_{ent} ensures that the student's predictions remain stable across domain distributions. With the introduction of $\mathcal{L}_{\text{kld}}$ in the ST stage, both entropy regularization and feature-level consistency are jointly optimized to promote stability. As shown in Fig. 8(b), higher values of λ_{ent} generally lead to a more stable feature space, but excessive regularization can cause oversmoothing, limiting the model's ability to adapt to

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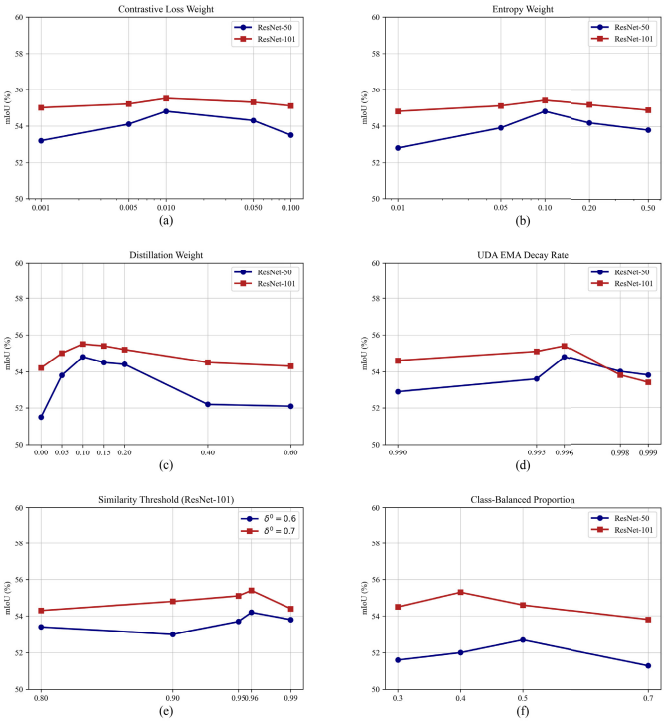


Fig. 8. Parameter sensitivity analysis of the proposed framework. (a) Loss weight λ_{inst}. (b) Loss weight λ_{ent}. (c) Temperature δ_{cal}. (d) EMA coefficient α. (e) Similarity threshold δ^0 and $\delta^{\max}$. (f) p_{ratio}.

TABLE V
PERFORMANCE COMPARISON UNDER DIFFERENT IMAGING CONDITIONS ON THE ISPRS DATASET

Tasks	Methods	mIoU (%)	F1 (%)
P_RGB V_IRRG	Source only	29.38	45.12
	SJT-Net	52.83	66.41
	DAFormer	51.17	68.35
	Ours	**55.05**	**71.39**
P_IRRG V_IRRG	Source only	34.24	41.37
	SJT-Net	54.89	67.02
	DAFormer	53.06	69.75
	Ours	**55.17**	**70.58**

the target domain. We have found $\lambda_{\text{ent}} = 0.1$ to yield optimal performance, balancing stability and adaptability.

3) Pseudolabel Correction: We have introduced an anchor-guided pseudolabel generator for the optimized ST stage. The calibration hyperparameter δ_{cal} controls the sensitivity of the feature-based correction process. As shown in Fig. 8(c), we evaluated the model's performance by varying δ_{cal} within $[0.01, 1.0]$. The results indicate that larger values oversmooth discriminative features, while smaller values produce overly sharp distributions that filter useful target features. Thus, we select 0.1 as the optimal value.

4) Distillation Weight: We have introduced the teacher–student network to better establish the knowledge transfer of the adaptation. The result in Fig. 8(d) indicates larger distillation weight makes the model less capable of handling these feature-level distributions, whereas a smaller α provides inadequate target guidance and destabilizes the source-domain semantic knowledge. We have set $\alpha = 0.996$ for optimal adaptation.

5) Dynamic Similarity Threshold: As shown in Fig. 8(e), we evaluated $\delta^{\max}$ across $[0.8, 0.99]$ under different initial lower bounds $\delta^0 \in \{0.6, 0.7\}$. The results indicate that a higher upper bound will discard valid positive pairs due to domain shift. The final configurations for δ^0 are 0.7, and $\delta^{\max}$ is 0.9 to achieve optimal performance.

6) Class-Balanced Proportion: Fig. 8(f) evaluates the initial p_{ratio} within $\{0.3, 0.4, 0.5, 0.7\}$. Results indicate that overly high values of p_{ratio} cause error accumulation, while lower values discard valuable target features and slow the adaptation. Therefore, $p_{\text{ratio}} = 0.4$ is selected as the optimal starting point.

V. CONCLUSION

In RS semantic segmentation, adapting models to unlabeled target domains without relying on annotations remains a challenge due to domain shifts driven by varying sensor specifications and geographical factors. Traditional ST frameworks suffer from these shifts, which manifest as label noise that accumulates and reinforces incorrect predictions, ultimately degrading model generalization. To overcome these limitations, we propose St-CGN, a two-stage framework that decouples semantic foundation learning from target domain adaptation. Experiments on ISPRS and LoveDA demonstrate that St-CGN outperforms state-of-the-art methods, particularly in distinguishing spectrally similar categories. Future work will extend this feature-consistency framework to multimodal tasks to further enhance domain invariance.

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